
Computer Science and Engineering Department, SVNIT, Surat

M.Tech. I – Semester I

Computer Vision and Image Processing (CSDS119)

Lab Assignment 3

1 Introduction

Computer Vision (CV) is the field of computing concerned with enabling machines to obtain meaningful information from visual data such as images, videos, and three-dimensional scenes. Image Processing (IP), on the other hand, focuses on operations that transform or analyze images to improve, extract, or represent useful information.

While many introductory CV applications operate on two-dimensional images, several modern applications require understanding the three-dimensional structure of objects and environments. This leads to the broader area of **3D Vision**, where geometry, depth, spatial relationships, viewpoints, and scene structure become important.

A conventional 2D image represents information primarily using image coordinates:

$$(x, y) \tag{1}$$

A 3D representation additionally includes depth:

$$(x, y, z) \tag{2}$$

where z represents the spatial/depth dimension.

Therefore, 3D vision systems frequently require suitable methods for representing and visualizing objects in three-dimensional space.

The present practical exploration investigates two tools:

1. **Spline**
2. **Womp**

The objective is not to learn every feature of these tools. Instead, the experiments focus on understanding how they can contribute to a 3D vision workflow through practical creation, representation, visualization, and interaction.

2 Assignment Objective

The assigned task involves exploring two freely available tools related to Computer Vision, Image Processing, 3D Vision, or Medical Image Analysis. For each tool, the practical work should include:

- Understanding the purpose of the tool.
- Identifying one CV/IP-related application.
- Performing one small practical experiment.
- Showing the input and output of the experiment.
- Identifying and explaining the CV/IP concept involved.
- Recording an observation or limitation.

The work presented in this report follows these requirements by treating Spline and Womp as tools that can support a **3D Vision workflow**. The experiments are intentionally small and focused rather than attempting to document every capability of either software.

3 Background

3.1 From 2D Vision to 3D Vision

A digital image can be interpreted as a two-dimensional array of pixels. Each pixel has a location and associated intensity or colour information. However, an image does not directly provide complete spatial information about the real-world scene.

3D vision attempts to recover, represent, or communicate spatial information about objects and environments.

A simplified conceptual workflow can be represented as:

In a practical workflow, the 3D representation may be generated by a computer vision or 3D reconstruction algorithm and then passed to tools for further modeling, visualization, inspection, or interaction.

Spline and Womp are relevant primarily at this latter stage.

3.2 Spatial Visualization

Spatial visualization involves presenting three-dimensional information in a manner that allows a user to understand:

- Object shape
- Relative position
- Orientation
- Depth

- Camera viewpoint
- Scene structure

This concept is particularly relevant to Spline because the tool allows 3D scenes to be presented and explored interactively.

4 Overview of the Tools

4.1 Spline

Spline is a web-based 3D design and visualization environment that allows users to create 3D scenes, manipulate objects, control cameras and lighting, apply materials, and build interactive experiences.

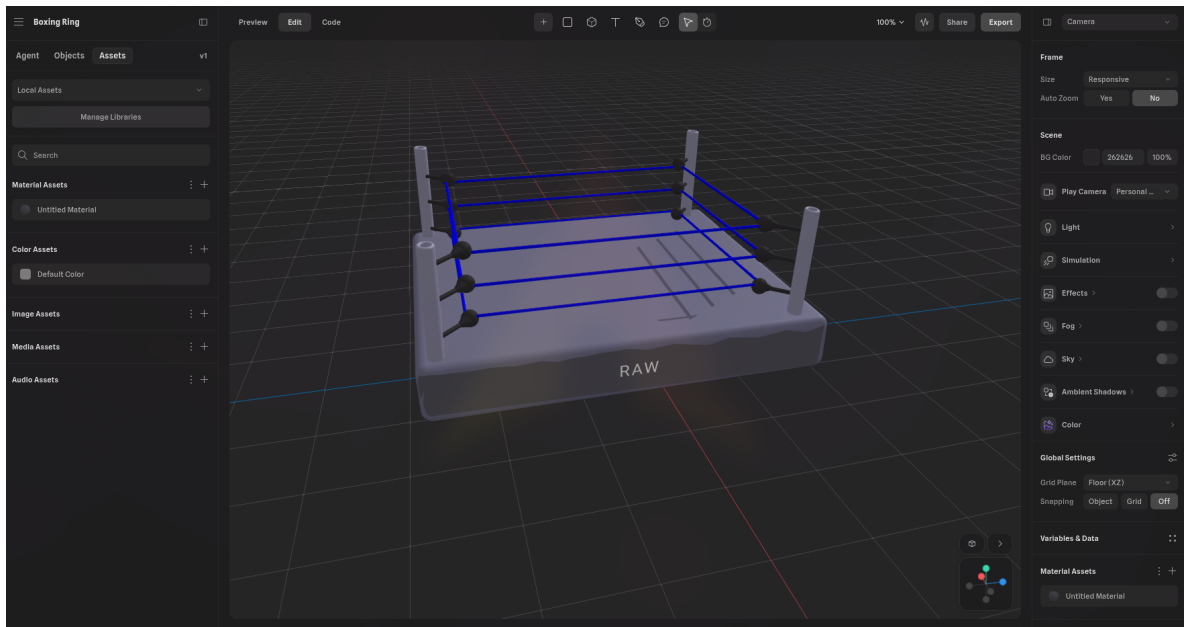


Figure 1: Spline experiment: 3D boxing ring scene created and visualized in the Spline workspace.

For the present assignment, its most relevant capabilities are:

- 3D object and scene creation
- Camera control
- Materials and lighting
- Animation
- Interaction

- Interactive scene visualization
- Web-based publishing

Its main relevance to the selected CV/IP concept is **interactive 3D visualization and spatial exploration**.

4.2 Womp

Womp is a browser-based 3D modeling environment designed for creating, modifying, and refining three-dimensional objects.

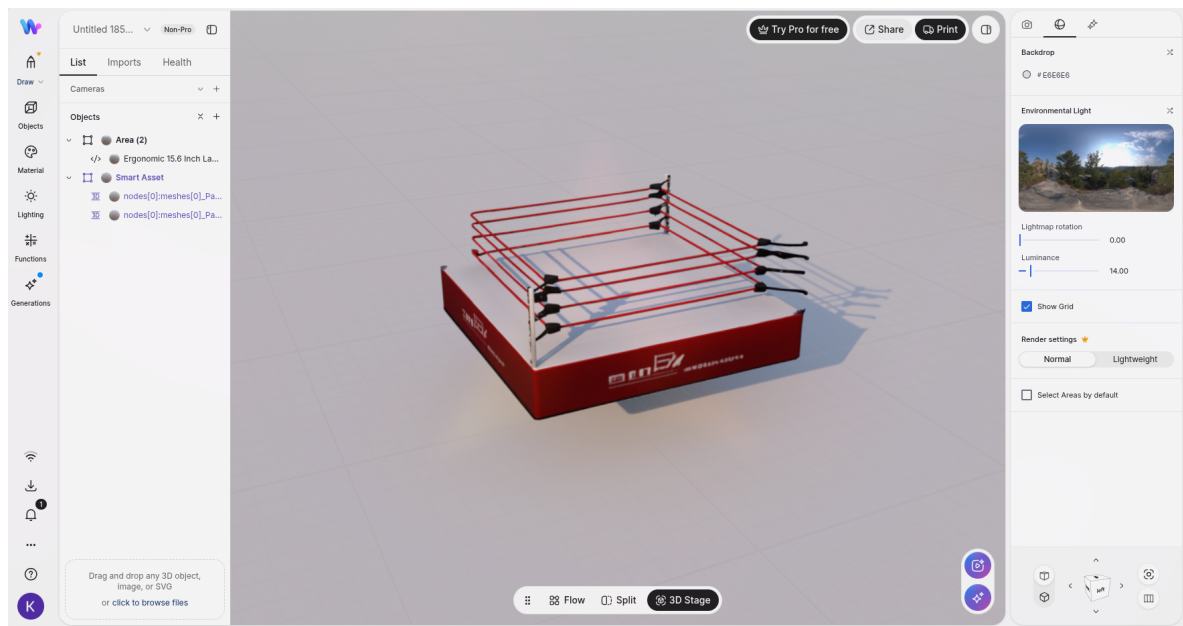


Figure 2: Womp experiment: 3D boxing ring scene created and visualized in the Womp workspace.

Relevant capabilities include:

- 3D object modeling
- Geometric shape manipulation
- Freeform modeling
- Materials and textures
- Lighting
- Asset-based modeling
- Import and export

- 3D printing-oriented workflows

Its primary relevance to the selected CV/IP concept is **3D object representation and geometry creation**.

5 Experiment 1: Spline

5.1 Experiment Objective

The objective of the Spline experiment was to create and visualize a three-dimensional boxing ring scene using basic geometric objects and scene controls.

The experiment was selected because a boxing ring naturally demonstrates multiple aspects of 3D scene representation:

- A base/platform represents the scene geometry.
- Cylindrical posts represent vertical structures.
- Ropes connect the posts spatially.
- Materials and colours provide visual information.
- Camera positioning determines the viewpoint.

The experiment therefore provides a simple example of **3D scene representation and spatial visualization**.

5.2 Input

The main inputs used in the experiment were:

- Basic geometric shapes
- Box geometry for the platform
- Cylindrical objects for the corner posts
- Rope-like objects for the ring boundaries
- Material and colour settings
- Lighting settings
- Camera settings

5.3 Experimental Procedure

The following procedure was followed.

5.3.1 Step 1: Create the Base

A box-shaped object was used as the basic platform of the boxing ring. Its dimensions were adjusted to provide a suitable rectangular base.

5.3.2 Step 2: Add Corner Posts

Four cylindrical objects were placed near the four corners of the platform. These objects represent the vertical corner posts of a boxing ring.

5.3.3 Step 3: Add Ropes

Multiple rope-like objects were positioned between the corner posts. This created the characteristic boundary structure of the ring.

5.3.4 Step 4: Configure the Scene

The scene was then adjusted using camera, lighting, and visual settings. These changes improved the visibility of the 3D structure and allowed the scene to be examined from a suitable viewpoint.

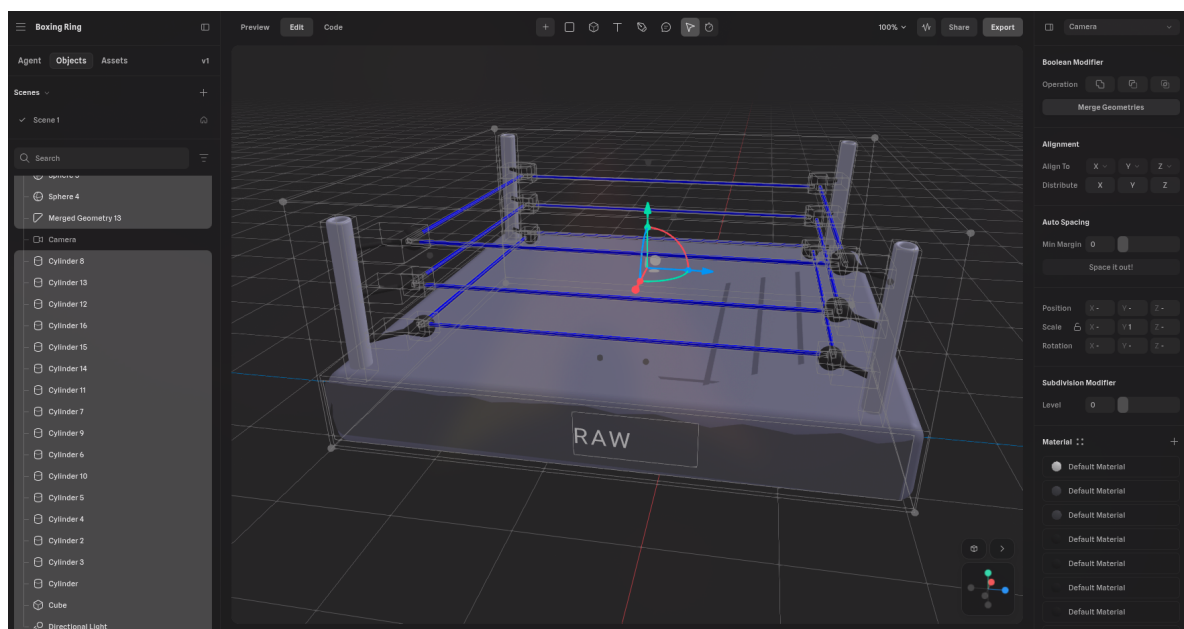


Figure 3: Spline experiment: 3D boxing ring scene visualized using different shapes.

5.4 Experimental Screenshot

Figure 13 shows the actual Spline workspace used during the experiment.



Figure 4: Spline experiment: 3D boxing ring front view.

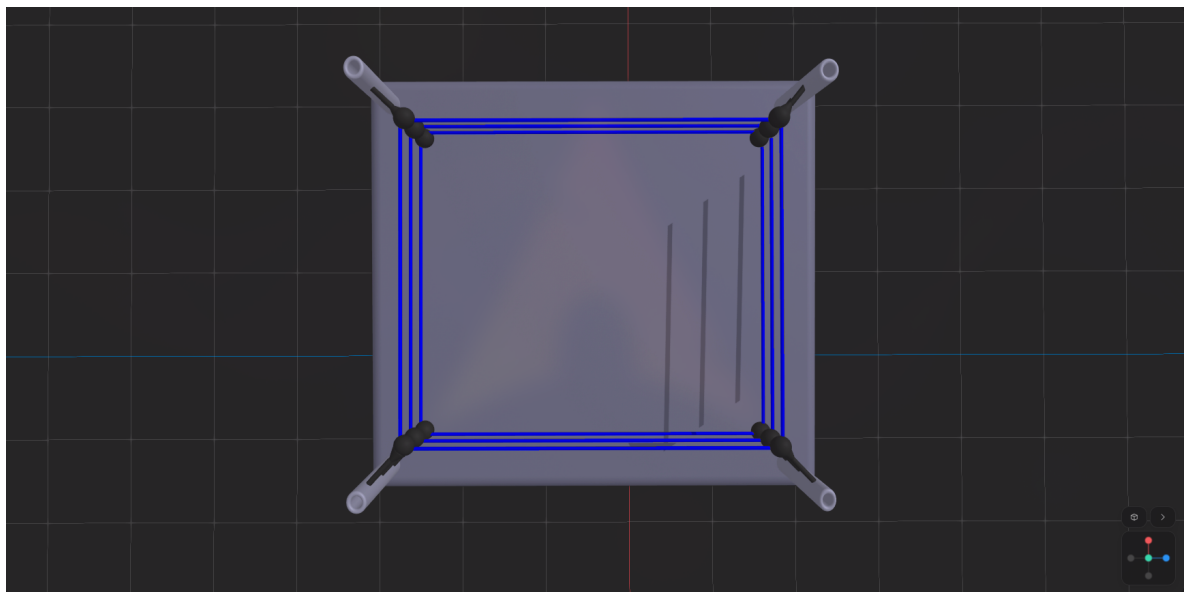


Figure 5: Spline experiment: 3D boxing ring scene top view.

5.5 Output

The output of the experiment was a 3D boxing ring scene containing:

- A rectangular platform
- Four corner posts

- Multiple ropes
- Applied visual/material properties
- A configured camera viewpoint

The resulting scene can be viewed as a spatial 3D representation rather than as a conventional 2D image.

5.6 CV/IP Concept

The primary CV/IP concept demonstrated by the experiment is:

3D Scene Representation and Spatial Visualization

The experiment demonstrates how multiple geometric components can be organized into a coherent three-dimensional scene.

Camera position also introduces the concept of **viewpoint**. Changing the camera changes the projection of the same 3D scene onto the user's screen.

Conceptually:

$$3D \text{ Scene} + \text{Camera Viewpoint} \rightarrow 2D \text{ Visual Projection} \quad (3)$$

This is an important relationship in computer vision because cameras observe 3D environments through 2D projections.

5.7 Observation

The main observations from the experiment were:

- Spline provides an intuitive environment for constructing and visualizing 3D scenes.
- Multiple objects can be arranged spatially to represent a larger real-world structure.
- Camera and lighting settings significantly influence how the 3D structure is perceived.
- Real-time visual feedback makes it easy to inspect changes to the scene.

5.8 Limitation

The major limitation in the context of this assignment is that Spline is primarily a 3D design and visualization environment rather than a dedicated computer vision processing framework.

It does not replace CV algorithms for tasks such as:

- Object detection
- Image segmentation
- Feature extraction
- Optical flow
- Depth estimation

Instead, it can be used to visualize or communicate 3D representations produced by such workflows.

6 Experiment 2: Womp

6.1 Experiment Objective

The objective of the Womp experiment was to create and refine a 3D object using geometric modeling, material selection, and lighting controls.

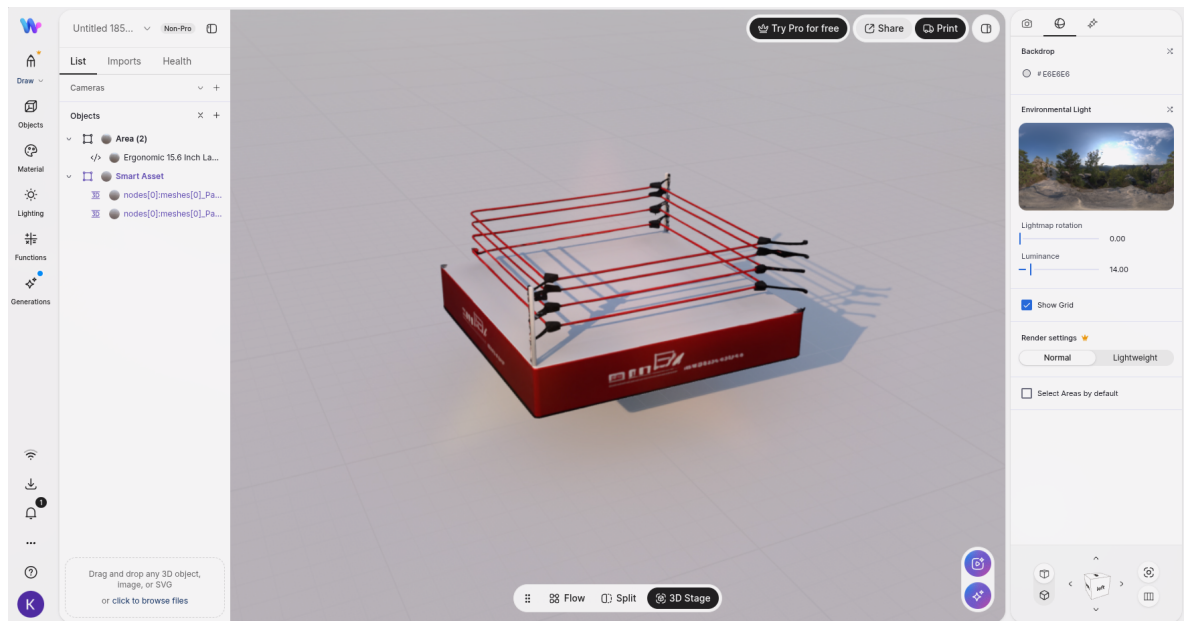


Figure 6: Womp experiment: 3D boxing ring scene created and visualized in the Womp workspace.

The experiment focuses on the representation of an object's geometry and appearance. This makes it suitable for demonstrating the concept of **3D object representation**.

6.2 Input

The experiment used the following components:

- Basic 3D geometry
- Geometric shapes and object structure
- Materials
- Surface appearance settings
- Environmental lighting
- Scene/render settings

6.3 Experimental Procedure

The experiment consisted of the following general stages.

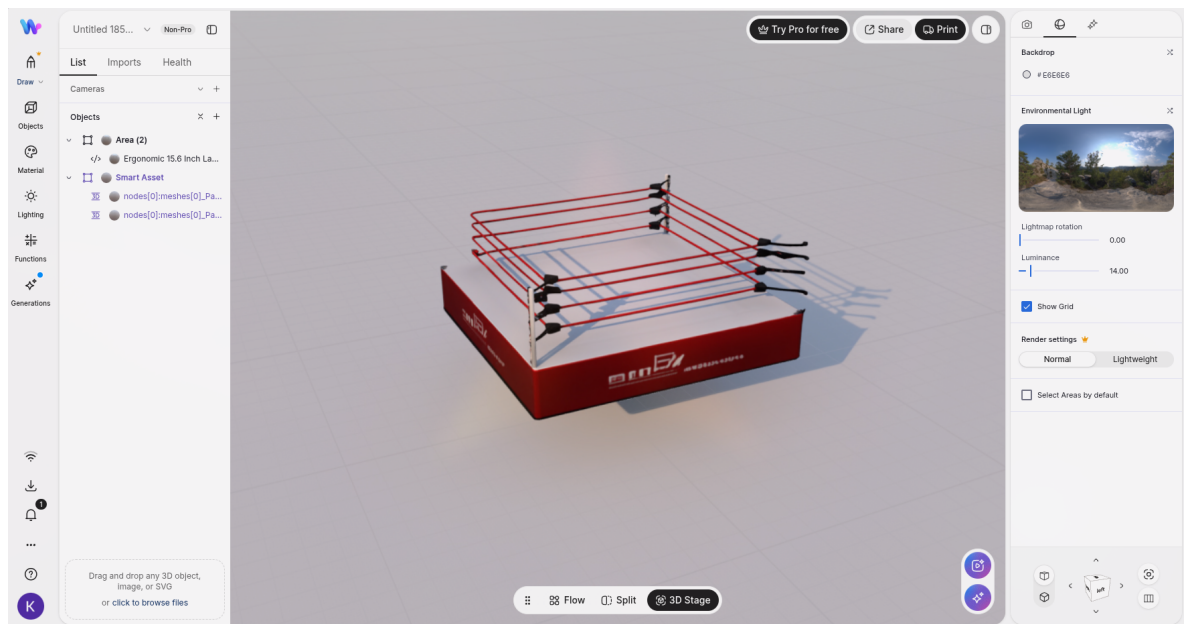


Figure 7: Womp experiment: 3D boxing ring scene front View.

6.3.1 Step 1: Create the Object Geometry

The required 3D object structure was created using Womp's modeling environment.

The geometry was adjusted to obtain the required overall form.

6.3.2 Step 2: Modify the Shape

Different geometric portions of the object were refined to obtain the desired shape.

This stage demonstrates the importance of geometry in a 3D representation.

6.3.3 Step 3: Apply Materials

Materials were selected and applied to the object.

This changes the visual appearance of the surface and helps distinguish different parts of the model.

6.3.4 Step 4: Adjust Lighting

Environmental lighting and rendering settings were adjusted to obtain a clearer visualization of the object.

6.4 Experimental Screenshot

Figure 8 shows the Womp workspace and the front view of the laptop stand created during the experiment. The model was designed as a 3D-printable laptop stand.

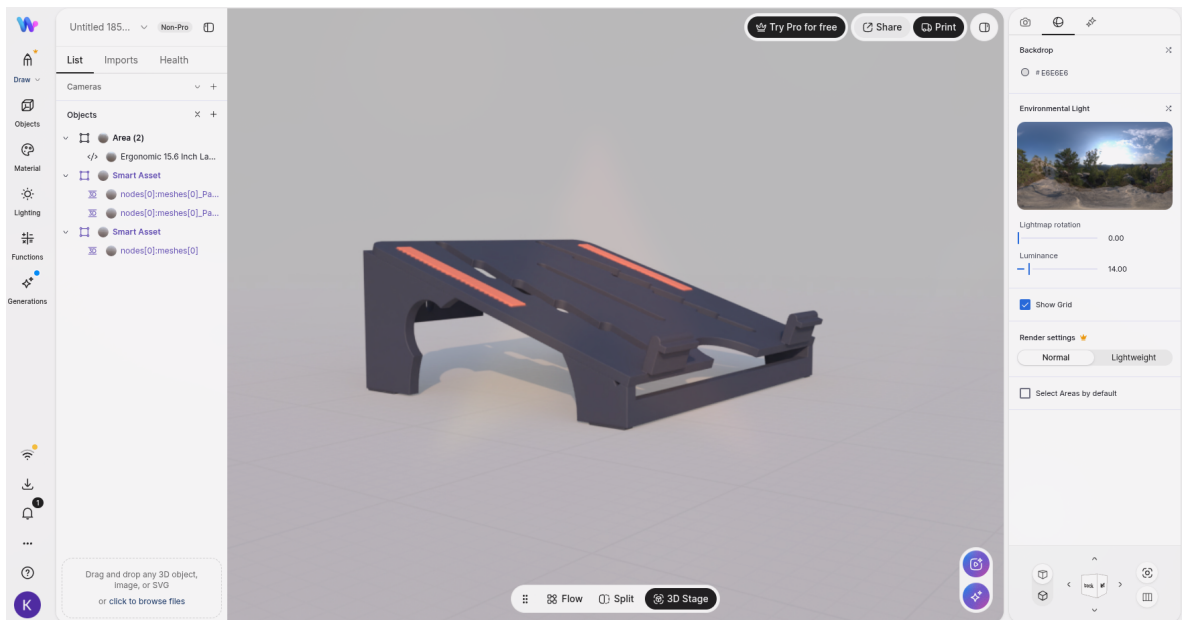


Figure 8: Womp experiment: Front view of the 3D laptop stand designed for 3D printing.

Figure 9 shows the top view of the same model, providing another viewpoint for examining its three-dimensional geometry and spatial structure.

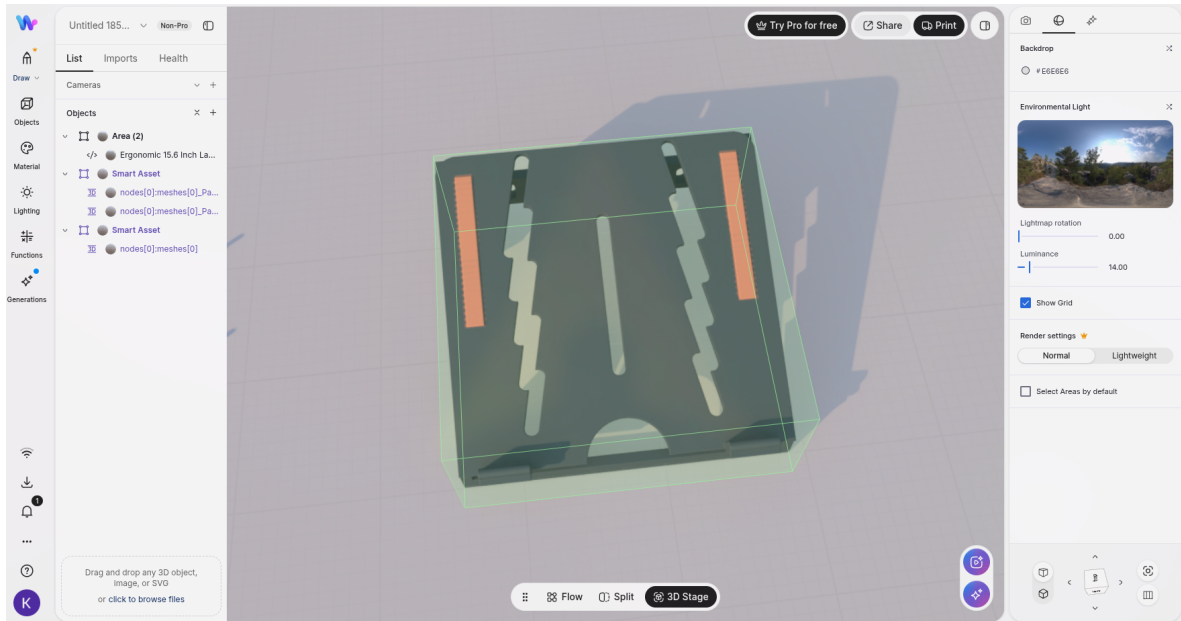


Figure 9: Womp experiment: Top view of the 3D laptop stand designed for 3D printing.

6.5 Output

The output was a 3D model of a laptop stand designed with the intention of being used for 3D printing. The resulting model contained:

- Three-dimensional geometric structure
- Inclined support surface for the laptop
- Stable supporting structure
- Defined surfaces and object dimensions
- Materials and visual appearance for model inspection

The resulting object is a manipulable 3D representation that can be further edited, inspected from different viewpoints, and prepared for export to a 3D-printing workflow.

6.6 CV/IP Concept

The primary concept demonstrated by the Womp experiment is:

3D Object Representation and Geometry

In 3D vision, an object can be represented using its geometric structure, spatial dimensions, surfaces, and relative positions in three-dimensional space.

The laptop stand experiment demonstrates how a real-world object can be represented digitally using 3D geometry. Different viewpoints can then be used to inspect the spatial structure of the same object.

The experiment illustrates the relationship:

$$\text{Object Design} \rightarrow \text{3D Geometric Representation} \rightarrow \text{Viewpoint-based Visualization} \rightarrow \text{3D Printing Workflow} \quad (4)$$

The geometry represents the physical structure of the laptop stand, while materials and lighting provide additional visual information for inspecting the model.

6.7 Observation

The main observations were:

- Womp provides a direct workflow for creating and modifying three-dimensional object geometry.
- The laptop stand could be represented as a complete 3D model rather than only as a two-dimensional drawing.
- Viewing the model from different viewpoints makes different geometric features and surfaces visible.
- The visual appearance of the model can be modified using materials and lighting without fundamentally changing its geometric structure.
- For a 3D-printing application, the geometric structure of the model is more important than its visual appearance alone. Factors such as dimensions, thickness, connected geometry, and print orientation need to be considered for successful fabrication.

6.8 Limitation

Womp is primarily a 3D modeling environment and is not a conventional computer vision analysis framework.

For example, it should not be considered a replacement for algorithms that automatically perform:

- Object detection
- Image classification and segmentation
- Feature matching

- Depth estimation
- Image-based 3D reconstruction

The laptop stand in this experiment was created manually as a 3D model; it was not reconstructed automatically from images or camera data.

Furthermore, creating a visually correct 3D model does not by itself guarantee successful 3D printing. Additional considerations such as dimensions, wall thickness, model geometry, print orientation, support structures, and printer/material constraints may need to be checked before fabrication.

Its relevance to CV/IP is therefore primarily through **3D object representation, geometric modeling, visualization, and downstream use of 3D information.**

7 Comparative Analysis of Spline and Womp

Although both Spline and Womp operate in the 3D domain, their primary purposes and practical roles are different. Based on the experiments performed, Spline was explored for interactive 3D scene visualization, whereas Womp was used to create a 3D model of a laptop stand intended for 3D printing.

The most important distinction observed during the practical exploration is:

Spline focuses on the 3D scene, while Womp focuses on the 3D object.

The comparison shows that the two tools can be viewed as serving different stages or purposes within a broader 3D workflow. Spline emphasizes how a 3D scene is presented and explored, while Womp emphasizes the creation and refinement of the geometry of a physical object.

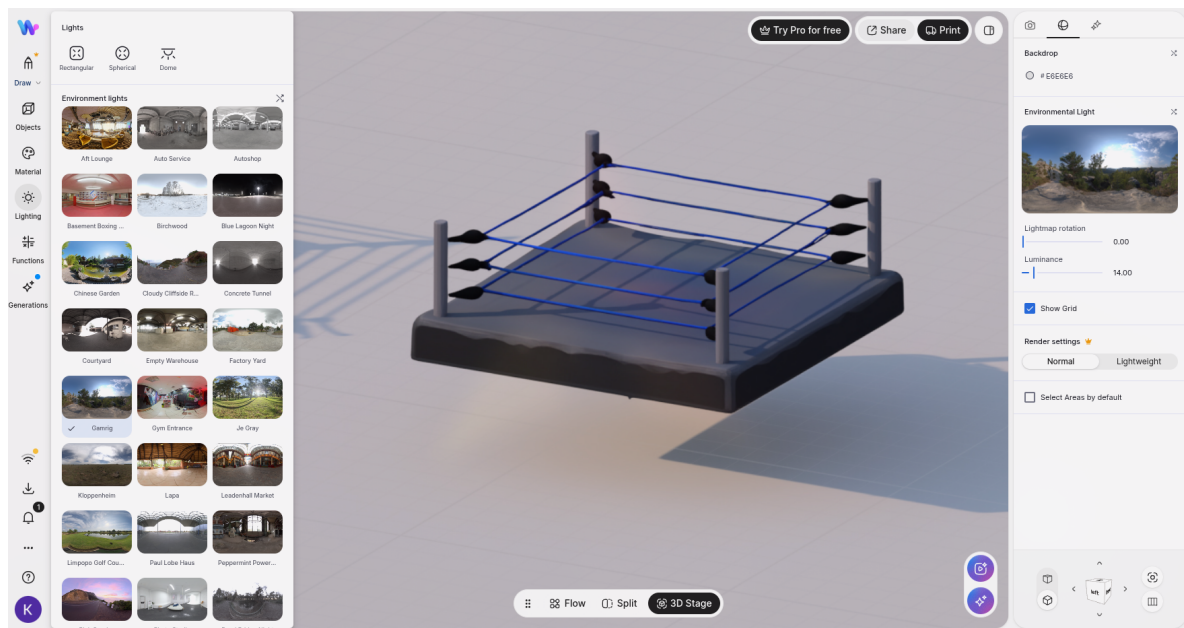
In the Womp experiment, the laptop stand was designed as a three-dimensional geometric model with the intention of producing a physical object through 3D printing. Therefore, the experiment provided a direct connection between **digital 3D object representation** and **physical fabrication**.

In contrast, the Spline experiment focused on representing and exploring a 3D scene interactively. Thus, the two experiments demonstrate different aspects of working with three-dimensional visual information: **Womp for object creation and fabrication-oriented modeling, and Spline for interactive visualization and scene exploration.**

**Womp creates and modifies the 3D object,
while Spline visualizes and brings the 3D scene to life.**

Table 1: Comparison of Spline and Womp Based on the Experiments

Aspect	Spline	Womp
Primary Focus	Interactive 3D visualization and scene experiences	3D modeling and object design for fabrication
Main Strength	Scene visualization, camera control, animation, and interaction	Creating, modifying, and refining 3D object geometry
Practical Experiment	Creation and visualization of an interactive 3D scene	Design of a laptop stand as a 3D-printable object
Object Handling	Objects are arranged and presented as components of an interactive scene	Objects are created and modified as three-dimensional geometric models
Visualization	Strong emphasis on interactive scene presentation and viewpoint exploration	Used to inspect the geometry, structure, and appearance of the modeled object
Camera / Viewpoint	Important for exploring the 3D scene from different viewpoints	Used to inspect the laptop stand geometry from different views
Materials	Used to enhance the appearance and presentation of the 3D scene	Used to visualize and refine the appearance of the laptop stand
Output	Interactive 3D scene / visual experience	3D model of a laptop stand intended for a 3D-printing workflow
CV/IP Relevance	3D scene representation and spatial visualization	3D object representation, geometric modeling, and physical fabrication

**Figure 10:** Womp experiment: 3D boxing ring scene imported from spline.

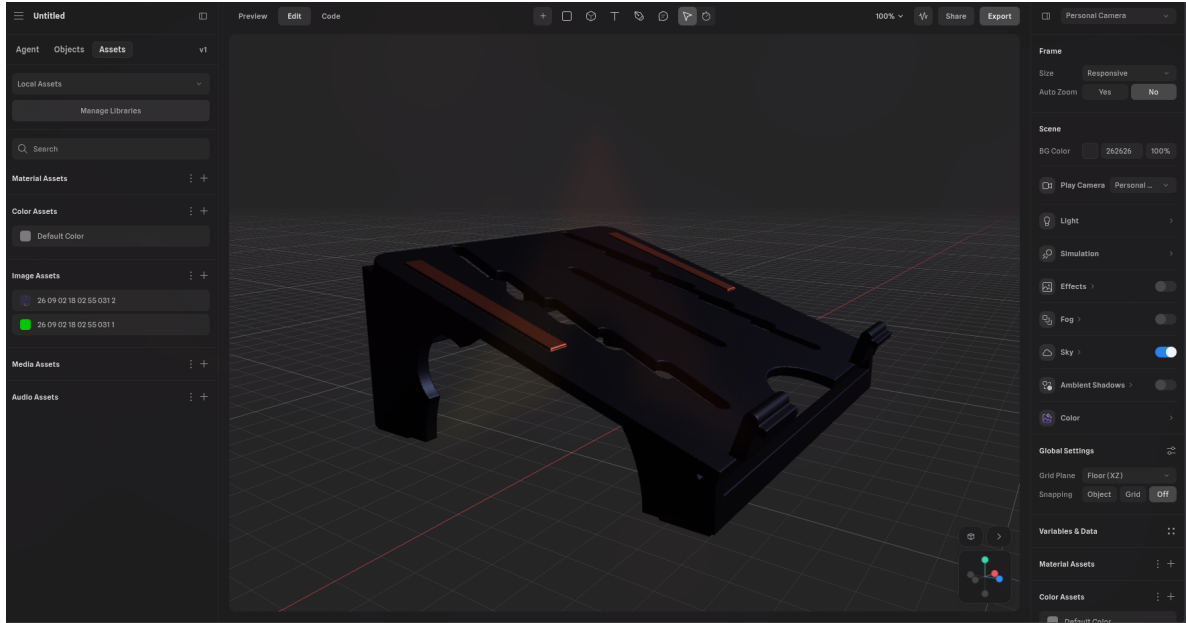


Figure 11: Spline experiment: 3D Laptop stand Imported from Womp.

7.1 Similarities

The tools also share several characteristics:

1. Both work with three-dimensional objects.
2. Both support visual modification of 3D content.
3. Both provide material/appearance-related functionality.
4. Both can be used through browser-based workflows.
5. Both can contribute to visualization-oriented 3D workflows.

7.2 Major Difference

The difference can be summarized as:



Figure 12: Spline experiment: 3D boxing ring scene created and visualized in the Spline workspace.

8 Integrated 3D Vision Workflow

The two tools can conceptually be positioned at different stages of a larger 3D vision workflow.

A simplified workflow is:

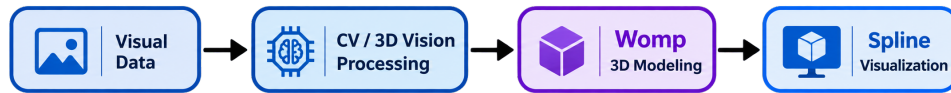


Figure 13: Spline experiment: 3D boxing ring scene created and visualized in the Spline workspace.

This diagram should not be interpreted as saying that Womp or Spline performs the CV processing stage. Instead, it illustrates how such tools can be positioned around a 3D vision workflow.

For example, a computer vision system may generate or estimate a 3D representation. That representation could then be edited, refined, visualized, or presented using appropriate 3D software.

9 Applications in CV/IP and 3D Vision

9.1 Applications of Spline

Spline can support several visualization-oriented applications:

- **Interactive 3D Visualization:** Presenting 3D objects and scenes that users can explore.
- **3D Scene Presentation:** Communicating spatial relationships between objects.
- **Product Visualization:** Presenting products as interactive 3D objects.
- **Scientific and Technical Visualization:** Representing technical concepts using three-dimensional scenes.
- **Educational Visualization:** Helping users understand spatial structures interactively.

- **Web-based 3D Experiences:** Publishing interactive 3D scenes for web-based exploration.

9.2 Applications of Womp

Womp can support applications where the primary requirement is creating or refining a 3D object:

- **3D Modeling**
- **Product Prototyping**
- **3D Printing**
- **3D Asset Creation**
- **Custom Product Design**
- **Character and Avatar Design**
- **Architectural and Organic Modeling**
- **3D Object Refinement**

10 CV/IP Concepts Demonstrated

10.1 3D Object Representation

The Womp experiment demonstrates how an object can be represented geometrically in three-dimensional space.

Instead of representing the object only as a collection of pixels, its shape is represented through 3D geometry.

This distinction is important because a 3D representation can preserve spatial relationships that are difficult to infer directly from a single 2D image.

10.2 3D Scene Representation

The Spline experiment demonstrates how several independent objects can be combined into a structured 3D scene.

For example:

$$\text{Platform} + \text{Posts} + \text{Ropes} = \text{Boxing Ring Scene} \quad (5)$$

The individual components have different positions, dimensions, orientations, and relationships.

10.3 Viewpoint and Projection

A 3D scene can be observed from multiple viewpoints.

A simplified camera model can be represented as:

$$P = K[R|t] \quad (6)$$

where:

- P is the camera projection matrix,
- K represents intrinsic camera parameters,
- R represents camera rotation,
- t represents camera translation.

The practical Spline experiment does not implement this camera model mathematically, but changing the camera viewpoint provides an intuitive visual demonstration of the underlying concept.

11 Experiment Comparison

Table 2: Comparison of the Practical Experiments

Parameter	Spline Experiment	Womp Experiment
Experiment	3D boxing ring scene	3D object modeling and refinement
Input	Basic 3D objects, ropes, camera and scene settings	3D geometry, materials and lighting settings
Main Operation	Scene creation and visualization	Object creation and geometry refinement
Primary Concept	3D scene representation and spatial visualization	3D object representation and geometry
Output	Interactive/visualized 3D scene	Refined 3D object model
Main Observation	Easy exploration and visualization of spatial scenes	Convenient creation and modification of 3D geometry
Main Limitation	Not a dedicated CV processing framework	Not a dedicated CV analysis framework

11.1 Geometry and Appearance

A 3D object has at least two important aspects:

1. **Geometry** – shape and spatial structure.
2. **Appearance** – material, colour, lighting, and texture.

Womp provides a direct environment for experimenting with both aspects.

12 Overall Observations

The practical exploration produced the following observations.

12.1 Observation 1: Different Primary Roles

The most important observation is that both tools operate in the 3D domain but emphasize different stages of working with 3D content.

Womp is more strongly oriented toward the object itself, whereas Spline is more strongly oriented toward the scene and its interactive presentation.

12.2 Observation 2: Visual Feedback

Both tools provide immediate visual feedback.

This is particularly useful during 3D modeling because small changes in geometry, material, lighting, or camera position can be observed directly.

12.3 Observation 3: Importance of Camera and Lighting

The same geometric object can appear substantially different depending on its viewpoint and lighting.

This demonstrates an important practical aspect of visual computing: the observed appearance of an object is affected not only by its geometry but also by how the scene is viewed and illuminated.

12.4 Observation 4: 3D Representation is More Than Shape

The experiments show that a 3D representation is not limited to geometry.

A practical scene may contain:

Geometry + Material + Lighting + Camera + Spatial Relationships

Together, these elements form a more complete visual representation.

13 Conclusion

This report presented the practical exploration of Spline and Womp in the context of Computer Vision, Image Processing, and 3D Vision.

The experiments demonstrated two complementary aspects of working with three-dimensional visual information.

The Womp experiment focused on **3D object modeling and geometry**. A 3D object was created and refined using geometric structures, materials, and lighting. This demonstrated how visual objects can be represented through manipulable three-dimensional geometry.

The Spline experiment focused on **3D scene representation and interactive visualization**. A boxing ring was constructed from multiple 3D components and visualized using camera and scene settings. This demonstrated how spatial relationships between objects can be communicated through an interactive 3D scene.

From a Computer Vision perspective, neither tool should be interpreted as a conventional CV algorithm. Instead, they provide useful environments for representing, refining, visualizing, and communicating 3D information around a broader computer vision or 3D vision workflow.

Overall, the practical exploration helped demonstrate that computer vision does not end with extracting information from images. Once spatial information is available, appropriate 3D tools can help represent, inspect, visualize, and communicate that information effectively.